

Programming concepts that start in XI and are examined in XII

One college period · 40–45 minutes · Programming bridge (Sindh XI C++)

What you will learn

- Know that BIEK examines C in Paper II, while the Sindh 2019 book uses C++ in XI
- Recall algorithm, flowchart, source vs object code, IDE
- Preview data types, control structures, arrays and functions

COMMON BOARD MISTAKE

Karachi Board Paper I (XI) is theory. Full C programs are Paper II. Still, Sindh Unit 5–6 expect you to practise C++ in the lab in Year 1. This lecture is the bridge; Class XII lectures teach C in exam style.

Algorithm. A step-by-step finite method to solve a problem. Pseudocode is the algorithm in English-like lines. A flowchart uses standard symbols (terminal, process, input/output, decision).

IDE. Integrated Development Environment — editor + compiler + debugger in one window (Turbo C++, Dev-C++, VS Code). Shortcuts in Turbo C: F2 save, F9 compile, Ctrl+F9 run, Alt+F5 user screen.

C/C++ program structure: preprocessor (#include), global declarations, main(), statements, return. Tokens: keywords, identifiers, constants, operators, separators. C is case-sensitive: int is a keyword, Int is not.

Idea	C / C++ sketch
Input / output	scanf("%d", &n); printf("%d", n); or cin >> n; cout << n;
Selection	if, if-else, else-if, switch-case, nested if
Loop	for, while, do-while, nested loops; break, continue, goto
Function	return-type name(parameters) { body }
Array	int a[5] = {1,2,3,4,5};
String	char s[20] = "BIEK"; strlen, strcpy, strcat, strcmp
Structure	struct Employee { char name[20]; float salary; };

```
#include <stdio.h>
int main(void) {
    int m1 = 80, m2 = 70, m3 = 90;
    float avg = (m1 + m2 + m3) / 3.0f;
    if (avg ≥ 40) printf("Pass %.2f\n", avg);
    else printf("Fail\n");
    return 0;
}
```

Lab list from the Sindh curriculum (XI): calculator, marksheet, prime test, digits of a number, factors, star pattern, larger of two via function, sort array, search, matrix add/multiply, reverse a name with strlen, employee structure.

الخوارزم مسئلہ حل کرنے کے قدم ہیں۔ فلو چارٹ خاکہ ہے۔ IDE میں کوڈ لکھ کر چلاتے ہیں۔ BIEK کے پیپر نو میں C زبان آتی ہے — اگلی کتاب (XII) میں پوری مشق ہے۔

CHECK YOURSELF

1. F9 in Turbo C?

2. Is C case-sensitive?

Answers: 1. Compile · 2. Yes

MCQs

1. An IDE is used to:
 - a. Print books
 - b. Write and run programs
 - c. Scan photos only
 - d. Connect fibre
2. Which is a loop structure?
 - a. if
 - b. switch
 - c. for
 - d. goto only

Key: 1=b, 2=c

Short questions (3 marks each)

1. What is an IDE? Write a few shortcut keys.
2. Differentiate source code and object code.
3. Why is C called case-sensitive?

Model short answers

1. IDE combines editor, compiler and debugger. Turbo C: F2 save, F9 compile, Ctrl+F9 run, Alt+F5 output screen.
2. Source code is human-written (file.c). Object/machine code is the compiler output that the CPU runs.
3. Uppercase and lowercase letters are different: TOTAL and total are two identifiers; keywords must be lowercase (int, not INT).